

## Curriculum Vitae – Anton Kuttyrev

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### General information

ORCID: 0000-0001-5933-6455

### Education

2020 PhD in Geology and Mineral Sciences, defended 2019, awarded via the Dissertation Council at the V.S. Sobolev Institute of Geology and Mineralogy (SB RAS), 2020; research conducted at the Institute of Volcanology and Seismology (RAS), Petropavlovsk-Kamchatsky.

2017 Saint Petersburg Mining University, Russia. Specialist degree with Distinction in Applied Geology (5-year full-time program, broadly equivalent to a Master of Science).

### Academic appointments

Current Lecturer in Exploration Geology, Cardiff University Kazakhstan, Astana, Kazakhstan. Research and teaching in economic and exploration geology, including layered intrusions, chromitites, platinum-group elements, critical minerals, and isotope geochemistry. Current duties combine teaching and curriculum development in the geology programme with research in economic and exploration geology. This includes delivering Year 1 teaching, developing a new field course for the Kazakhstan programme, and building field-based learning into the curriculum. I teach World of Dynamic Environments (EA3504), a 20-credit Year 1 module that uses geological and environmental case studies to introduce students to dynamic Earth systems.

2025–2026 Postdoctoral Research Fellow, School of Geosciences, University of the Witwatersrand, Johannesburg, South Africa. Research on critical minerals in layered mafic-ultramafic intrusions and Bushveld chromitites within the DSTI CIMERA industry–public consortium, with a focus on Ni-Cu-PGE systems, chromitites, and the petrogenesis of the Critical Zone.

2023–2025 Research Associate, School of Earth and Environmental Sciences, Cardiff University, Wales, UK. Critical-minerals and PGE geochemistry in layered intrusions and related volcanic rocks, as part of the Horizon Europe SEMACRET consortium on energy-transition mineral systems, in an interdisciplinary team spanning geology, geophysics, and social sciences.

2022–2023 Fulbright Visiting Scholar, School of Earth Sciences, University of Oregon, USA. Investigation of Early Archean climate using triple oxygen and hydrogen isotopes in metalavas from the 3.8 Ga Saglek-Hebron terrane, including development of isotope fractionation models.

2020–2022 Research Associate, Institute of Volcanology and Seismology, Petropavlovsk-Kamchatsky, Russia. Research on PGE mineralization and hydrothermal processes in subduction-related magmatic systems in Kamchatka.

2017–2020 PhD candidate, Institute of Volcanology and Seismology, Petropavlovsk-Kamchatsky, Russia. Doctoral research on geology and platinum mineralization of concentrically zoned dunite-clinopyroxenite-gabbro complexes of the Tamanvayam and Epilchik groups.

2012–2017 Engineer (0.5 FTE), Karpinsky Russian Geological Institute, Saint-Petersburg, Russia. Work on regional geology and petrology, including petrography and geochemical data for igneous and metamorphic rocks.

### Industry appointments

2018 Geologist (summer fieldwork), Kirganik Ltd. (Amur Minerals subsidiary), Kamchatka, Russia. Exploration for porphyry Cu and epithermal Au systems, including field mapping, alteration and mineralization logging, and geochemical sampling.

2016–2017 Geologist (summer fieldworks), Kamp Ltd., Kamchatka, Russia. Junior exploration for PGE mineralization, including prospecting, field mapping, heavy-mineral sampling, rock and soil geochemical sampling.

2015 Field technician (summer fieldwork), Kamchatgeologia, Kamchatka, Russia. Regional geological mapping at 1:200 000 scale, including traverses, outcrop description, and systematic sampling for petrography and geochemistry.

### **Research focus**

Petrology and mineralogy of mafic-ultramafic and island-arc systems; melt-rock and fluid-rock interaction; origin of chromitites; PGE mineralogy from hand-sample to nanoscale, including alloys and PGE oxides; PGE/PGM cycling and mobility in melts and fluids in ultramafic rocks and layered intrusions; oxygen isotopes as tracers of crustal recycling and early Earth surface environments; critical minerals in the context of the energy transition.

### **Teaching experience**

Current teaching: Cardiff University Kazakhstan. Lecturer for World of Dynamic Environments (EA3504), a 20-credit Year 1 module on dynamic Earth systems, taught through geological and environmental case studies.

Current development: New field course development. Designing and building a new field course from scratch for the Kazakhstan programme, including its academic structure, field activities, and teaching materials.

University of the Witwatersrand: 5-lecture module on magmatic mineral deposits for honours students, demonstrator for the honours field trip (Northern Bushveld)

Cardiff University: practical sessions for the mineral deposits course; guest lectures on mineral deposits; demonstrator for field courses in areas of geological mapping, exploration geology and environmental geoscience (courses in Cyprus, Cornwall, and South Wales)

University of Oregon: demonstrator for petrology field courses; guest lectures in petrology and geohazards

### **Formal teaching training**

Teaching to Improve the Quality of Student Learning, Cardiff University, 2024

Lecturing Skills, Cardiff University, 2024

Being a Good LGBTQ+ Ally, Cardiff University, 2024

### **Student supervision**

2026–present Co-supervisor (together with prof. Roger Gibson) of the project “Calc-silicate xenoliths in the lower zone of the Bushveld Complex and their implications for platinum-group element mineralization” (student Mensacia Phaswane)

### **Additional professional training**

PetThermoTools virtual workshop, 2026

Reading the rocks, Polokwane, South Africa, 2025

Resource Geology Workshop at the 2025 GeoCongress, Bloemfontein, South Africa

MDSG workshop “Magmatic Ni-Cu Mineral Systems in Europe”, Edinburgh, 2024

TESCAN electron microscopy course, Saint-Petersburg, 2021

Gold Short Course, Prague, 2018

### **Analytical and modelling skills**

Hands-on: SEM-EDS, Raman spectroscopy, field portable XRF, clean-lab chemistry and mass spectrometry for Rb-Sr, Sm-Nd, triple-oxygen isotopes (laser fluorination line), LA-ICP-MS.

Data reduction and interpretation: EBSD, X-ray diffraction, synchrotron micro-XRF/XAS, FIB-TEM, EPMA, atom-probe tomography.

Modelling and data processing: Python and spreadsheet-based workflows, MELTS- and MAGEMin-based thermodynamic modelling, Monte Carlo modelling of triple-oxygen isotope exchange, mass-balance and mixing models.

### **Scientific service**

Minerals and Energy Research Centre, Cardiff University – coordinator (2024–2025)

2023 International Platinum Symposium, Cardiff, UK – member of organizing committee

Reviewer for: 'Ore Geology Reviews'; 'Mineralium Deposita'; 'Nature Communications'; 'Chemical Geology'; 'Journal of Petrology'; 'Australian Journal of Earth Sciences'; 'Contributions to Mineralogy and Petrology'; 'Journal of the Geological Society'; 'Lithos'; 'Marine Geology'.

### **Competitively awarded funding (converted from foreign currencies, approx. EUR)**

2024 In-kind analytical access through AuScope Access Program (Curtin University): "Silicate–oxide–platinum group minerals intergrowths: implications for unknown aspects of low-T PGE behaviour" - approx. EUR 10,700

2022–2023 Fulbright Scholarship: Fulbright Visiting Scholar - approx. EUR 26,600

2020–2022 RFBR Project No. 20-35-70008 (project executor): "Formation and alteration of minerals in active volcanic hydrothermal systems (Kamchatka)" - approx. EUR 114,000

2020–2021 RFBR Project No. 20-05-00290 (principal investigator): "Distribution of Ni-Fe-Co intermetallic compounds and platinum-group minerals in mafic and ultramafic rocks of various formational types" - approx. EUR 21,500

2018–2020 RSCI Project No. 16-17-10145 (project executor): "Melt-fluid evolution, magmatic immiscibility, and metal budgeting in subduction-related magmas" - approx. EUR 171,000

2018–2019 RFBR Project No. 18-35-00126 (principal investigator): "Typomorphism and formation conditions of PGE minerals in concentrically zoned ultramafic massifs (Epilchik area, Koryak-Kamchatka belt)" - approx. EUR 9,700

Total research funding received: approx. EUR 353,600

### **Field work**

2025–ongoing Mapping and sampling of the Critical Zone across western and eastern Bushveld Complex, South Africa (one month fieldwork + regular mine visits)

2023–2025 Fieldwork at the Rhiw layered intrusion, Wales, UK (8 days in total)

2021 Principal investigator and organizer of a helicopter-supported expedition (12 participants) to Tumrok Range arc picrites, Kamchatka (three weeks)

2020 Sampling of alluvial awaruite in Kamchatsky Mys ophiolite, Kamchatka (two weeks)

2020 Documenting cooling of lava flows and caves of 2012–13 Tolbachik Fissure Eruption (two weeks); observations of Klyuchevskoy volcano

2019 Underground and field sampling of mineralized intrusions and related flood basalts, Norilsk (two weeks)

2018 Documenting temperatures of lava in the caves of 2012–13 Tolbachik Fissure Eruption (two weeks); visual observations of Shiveluch volcano

2018 Fieldwork on Kamchatsky Mys ophiolite (two weeks)

2018 prospective works on Kirganik Cu porphyry and epithermal Au occurrence (one month)

2016–2017 Prospecting for lode and placer PGE mineralization, Epilchik and Matysken areas, Koryak Highlands (5 months in total)

2015 Geological mapping 1:200 000 scale, Koryak Highlands (3 months)

## Professional societies

Society of Geology Applied to Mineral Deposits (SGA), 2017–present

## Awards

2023 Minerals Young Investigator Award

2021 Marie Skłodowska-Curie Actions Seal of Excellence

## Public outreach

Experience of school visits in the UK and Russia covering ages 8 to 16: multiple interactive lectures about volcanoes, minerals, rocks, societal impact of geology, and career perspectives of a geoscientist.

## Publications

Published peer review articles:

1. **Kutyrev, A.**, Filimonova, O., Saxey, D.W., Vymazalová, A., Evans, K., Kamenetsky, V.S., Wermeille, D., Fougereuse, D., Reddy, S.M., 2026. Formation of natural IrO<sub>2</sub> through Os-Ir-Ru alloy oxidation. *Chemical Geology*, 123167. <https://doi.org/10.1016/j.chemgeo.2025.123167>.
2. Eslami, A., Chardelin, M., O'Driscoll, B., Malvoisin, B., Evans, K., Grieco, G., Bussolesi, M. & Kutyrev, A. (2026). Chromitite–mafic dike interfaces as archives of syn-kinematic fluid–rock interaction, chromium mobility and Ca-metasomatism, Nain Ophiolite, Central Iran. *Journal of Petrology*, egag076. <https://doi.org/10.1093/petrology/egag076>
3. **Kutyrev, A.**, Bindeman, I. N., O'Neil, J. & Rizo, H., 2024. Seawater-oceanic crust interaction constrained by triple oxygen and hydrogen isotopes in rocks from the Saglek-Hebron complex, NE Canada: Implications for moderately low- $\delta^{18}\text{O}$  Eoarchean ocean. *Chemical Geology* 670. <https://doi.org/10.1016/j.chemgeo.2024.122378>
4. **Kutyrev, A.**, Kamenetsky, V. S., Chayka, I. F., Nekrylov, N. A., Kryuchkova, L., Shilovskikh, V. V., Kontonikas-Charos, A., Krashenninnikov, S. P., Sapegina, A. & Perchuk, A. L., 2024. Metamorphic origin of large nuggets of platinum-group metals: evidence from multiphase inclusions in Os-Ir-Ru alloys from the Adamsfield placer, Tasmania. *Journal of Petrology*. 10.1093/petrology/egae086
5. **Kutyrev, A.V.**, Kamenetsky, V.S., Park, J.-W., Maas, R., Demonterova, E.I., Antsiferova, T.N., Ivanov, A.V., Hwang, J., Abersteiner, A., Ozerov, A.Y., 2021. Primitive high-K intraoceanic arc magmas of Eastern Kamchatka: Implications for Paleo-Pacific tectonics and magmatism in the Cretaceous. *Earth-Science Reviews* 220, 1–22. <https://doi.org/10.1016/j.earscirev.2021.103703>.
6. **Kutyrev, A.V.**, Kamenetsky, V.S., Sidorov, E.G., Abersteiner, A., Chubarov, V.M., 2020. Silicate inclusions in isoferroplatinum: Constraints on the origin of platinum mineralization in podiform chromitites. *Ore Geology Reviews* 119, 1–13. <https://doi.org/10.1016/j.oregeorev.2020.103367>.
7. **Kutyrev, A.**, Zelenski, M., Nekrylov, N., Savelyev, D., Kontonikas-Charos, A., Kamenetsky, V.S., 2021. Noble metals in arc basaltic magmas worldwide: a case study of modern and pre-historic lavas of the Tolbachik volcano. *Frontiers in Earth Science* 9. <https://doi.org/10.3389/feart.2021.791465>.
8. **Kutyrev, A.V.**, Sidorov, E.G., Kamenetsky, V.S., Chubarov, V.M., Chayka, I.F., Abersteiner, A., 2021. Platinum mineralization and geochemistry of the Matysken zoned Ural-Alaskan type complex and related placer (Far East Russia). *Ore Geology Reviews* 130, 1–17. <https://doi.org/10.1016/j.oregeorev.2020.103947>.
9. **Kutyrev, A.**, McDonald, A., Tamura, N., Cabri, L.J., Tolstykh, N., Pekov, I. 2023. Sidorovite, PtFe<sub>3</sub>, a new mineral from the Snegovaya River placer, Koryak Highlands, Russia. *The Canadian Journal of Mineralogy and Petrology*, 61(5): 1021–1030. <https://doi.org/10.3749/2200072>
10. **Kutyrev, A.**, Kamenetsky, V. S., Kontonikas-Charos, A., Savelyev, D. P., Yakich, T. Y., Belousov, I. A., et al., 2023. Behavior of platinum-group elements during hydrous metamorphism: constraints from awaruite (Ni<sub>3</sub>Fe) mineralization. *Lithosphere*, 2023(1). [https://doi.org/10.2113/2023/lithosphere\\_2023\\_126](https://doi.org/10.2113/2023/lithosphere_2023_126)
11. **Kutyrev, A.V.**, Zhirnova, T.S., 2019. Concentrically-zoned massifs of the Tamanvayam group (Koryak-

Kamchatka Platiniferous Belt): structure, age, petrological and geochemical aspects. Russian Journal of Pacific Geology 13, 350–363. <https://doi.org/10.1134/S1819714019040031>.

12. **Kutyrev, A.V.**, Sidorov, E.G., Antonov, A.V., Chubarov, V.M., 2018. Platinum-group mineral assemblage of the Prizhimny Creek (Koryak Highland). Russian Geology and Geophysics 59, 935–944. <https://doi.org/10.1016/j.rgg.2018.07.014>.

13. Rhoads, E. A., **Kutyrev, A.**, Bindeman, I. N., Lynn, K. J., Trusdell, F. A., Downs, D. T., Edwards, H. R., Cook, G. W. & Day, J. M. D., 2025. Rhenium-osmium and oxygen isotope homogeneity during the 2022 Mauna Loa eruption and implications for basaltic magma storage. Bulletin of Volcanology 87. <https://doi.org/10.1007/s00445-025-01825-0>

14. Chayka, I. F., Kamenetsky, V. S., **Kutyrev, A. V.**, Barnes, S. J., Stepanov, A. S., Konstantinov, S. E., Baykov, N. I., Kryazhev, S. G., Belousov, I. A., Zhukova, I. A., Scherbakov, V. D., Kalugin, V. M., Ivanov, A. V., Bryanskiy, N. V. & Park, J. W., 2025. Erratic behavior of platinum-group elements along an olivine-Cr-spinel cotectic in sulfide-undersaturated basaltic magma: roles of Cr-spinel and alloys. Sci Rep 15, 14366. <https://doi.org/10.1038/s41598-025-97281-z>

15. Yakubovich, O., **Kutyrev, A.**, Sidorov, E., Travin, A., 2022.  $^{190}\text{Pt}$ - $^4\text{He}$  dating of platinum mineralization in Ural-Alaskan-type complexes in the Kamchatka region: evidence for remobilization of platinum-group elements. Mineralium Deposita. <https://doi.org/10.1007/s00126-022-01103-5>.

16. Bukhanova, D.S., **Kutyrev, A.V.**, Sidorov, E.G., Chubarov, V.M., 2022. First finding of platinum group minerals at the Malmyzh porphyry gold-copper deposit, Khabarovsk Krai, Russia. Geology of Ore Deposits 63, 675–681. <https://doi.org/10.1134/s1075701521070035>.

17. Sidorov, E.G., **Kutyrev, A.V.**, Zhitova, E.S., Agakhanov, A.A., Sandimirova, E.I., Vymazalova, A., Chubarov, V.M., Zolotarev, A.A., 2021. Kufahrite, PtPb, a new mineral from Ledyanoy Creek placer, Galmoenan ultramafic complex, Koryak Highlands, Russia. Mineralogical Magazine 85, 254–261. <https://doi.org/10.1180/mgm.2021.18>.

18. Stepanov, S.Y., **Kutyrev, A.V.**, Lepekhina, E.N., Sharpenok, L.N., Antonov, A.V., Kutyreva, M.E., 2021. Age of the dike complex in the dunite “core” of the Kamenushinsky clinopyroxenite–dunite massif, Ural Platinum belt, Middle Urals. Geochemistry International 59, 559–576. <https://doi.org/10.1134/s0016702921060094>.

19. Sidorov, E., **Kutyrev, A.**, Chubarov, V., Zhitova, E., 2021. Platinum mineralization of the Epilchik Ural-Alaskan type zoned complex (Far East Russia). Mineralium Deposita 56, 143–160. <https://doi.org/10.1007/s00126-020-01008-1>.

20. Sidorov, E.G., **Kutyrev, A.V.**, Zhitova, E.S., Chubarov, V.M., Khanin, D.A., 2019. Origin of platinum-group mineral assemblages from placers in rivers draining from the Ural-Alaskan type Itchayvayamsky ultramafics, Far East Russia. Canadian Mineralogist 57, 91–104. <https://doi.org/10.3749/canmin.1800040>.

21. Chayka, I.F., Izokh, A.E., Kamenetsky, V.S., Sokol, E.V., Lobastov, B.M., Kontonikas-Charos, A., Zelenski, M.E., **Kutyrev, A.V.**, Sluzhenikin, S.F., Zhitova, L.M., Shvedov, G.I., Shevko, A.Y., Gora, M.P., 2023. Origin of chromitites in the Norilsk-1 intrusion (Siberian LIP) triggered by assimilation of argillaceous rocks by Cr-rich basic magma. Lithos 454–455. <https://doi.org/10.1016/j.lithos.2023.107254>.

22. Zhitova, E.S., Nuzhdaev, A.A., Davydova, V.O., Sheveleva, R.M., Zhegunov, P.S., Kuznetsov, R.A., **Kutyrev, A.V.**, Khokhlova, M.A., Vlasenko, N.S. (2025). Calcareous skarn-like mineral paragenesis from unaltered basalt of the Alaid volcano (Kuril–Kamchatka island arc). Minerals 15. <https://doi.org/10.3390/min15030237>

23. Pekov, I.V., Zubkova, N.V., Yapaskurt, V.O., Belakovskiy, D.I., Britvin, S.N., Agakhanov, A.A., Turchkova, A.G., Sidorov, E.G., **Kutyrev, A.V.**, Blatov, V.A., Pushcharovsky, D.Y., 2022. Nishanbaevite,  $\text{KAl}_2\text{O}(\text{AsO}_4)(\text{SO}_4)$ , a new As/S-ordered arsenate-sulfate mineral of fumarolic origin. Mineralogy and Petrology. <https://doi.org/10.1007/s00710-022-00803-0>

24. Stepanov, S., Palamarchuk, R., **Kutyrev, A.**, Lepekhina, E., Sharpenok, L., Shagalov, E., Minervina, E., 2024. Nature of perovskite mineralization of silicate-carbonate veins in the margins of Kusinko-Kopanskaya layered intrusion (South Urals, Russia). Minerals 14, 478.

<https://doi.org/10.3390/min14050478>

25. Eslami A., Grieco G., Bussolesi M., Ichiyama Y., Lenaz D., Skogby H., **Kutyrev A. V.**, Cavallo A. and Khedr M. Z. (2023) Co-occurrence of compositionally variable chromitites in the Sabzevar ophiolite, NE Iran. *Lithos* 446-447. <https://doi.org/10.1016/j.lithos.2023.107133>
26. Stepanov, S.Y., Palamarchuk, R.S., **Kutyrev, A.V.**, Shilovskikh, V.V., Petrov, S.V., 2023. Mineralogy and geochemistry of platinum-group elements in the zoned mafic-ultramafic intrusions of the Uralian Platinum belt, Russia. *Journal of Geochemical Exploration* 246. <https://doi.org/10.1016/j.gexplo.2023.107155>
27. Chayka, I.F., Baykov, N.I., Kamenetsky, V.S., **Kutyrev, A.V.**, Pushkarev, E.V., Abersteiner, A., Shcherbakov, V.D., 2023. Volcano-plutonic complex of the Tumrok Range (Eastern Kamchatka): an example of the Ural-Alaskan type Intrusion and related volcanic series. *Minerals* 13, 126.
28. Pekov I.V., Zubkova N.V., Agakhanov A.A., Yapaskurt V.O., Belakovskiy D.I., Britvin S.N., Sidorov, E.G., **Kutyrev, AV**, Pushcharovsky, D.Yu. New arsenate minerals from the Arsenatnaya fumarole, Tolbachik volcano, Kamchatka, Russia. XIX. Axelite,  $\text{Na}_{14}\text{Cu}_7(\text{AsO}_4)_8\text{F}_2\text{Cl}_2$ . *Mineralogical Magazine*. 2022:1-25.
29. Cui, M.-M., Su, B.-X., Wang, J., Wu, Y., **Kutyrev, A.**, 2022. Early Paleozoic subduction fingerprints of the Paleo-Asian ocean in easternmost Central Asian Orogenic Belt (CAOB): Identification of the oldest Alaskan-type complex in the CAOB. *GSA Bulletin*. <https://doi.org/10.1130/b36307.1>.
30. Nekrylov, N., Kamenetsky, V.S., Savelyev, D.P., Gorbach, N.V., Kontonikas-Charos, A., Palesskii, S.V., Shcherbakov, V.D., **Kutyrev, A.V.**, Savelyeva, O.L., Korneeva, A.A., Kozmenko, O.A., Zelenski, M.E., 2022. Platinum-group elements in Late Quaternary high-Mg basalts of eastern Kamchatka: Evidence for minor cryptic sulfide fractionation in primitive arc magmas. *LITHOS*. <https://doi.org/10.1016/j.lithos.2022.106608>.
31. Ismagilova, R.M., Zhitova, E.S., Krivovichev, S.V., Sergeeva, A.V., Nuzhdaev, A.A., Anikin, L.P., Krzhizhanovskaya, M.G., Nazarova, M.A., Kupchenenko, A.N., Zolotarev, A.A., **Kutyrev, A.V.**, Bukhanova, D.S., Kuznetsov, R.A., Khanin, D.A., 2021. Phase evolution from volborthite,  $\text{Cu}_3(\text{V}_2\text{O}_7)(\text{OH})_2 \cdot 2\text{H}_2\text{O}$ , upon heat treatment. *Minerals* 11, 1312. <https://doi.org/10.3390/min11121312>
32. Savelyev, D.E., Shilovskikh, V.V., Sergeev, S.N., **Kutyrev, A.V.**, 2021. Chromian spinel neomineralisations and the microstructure of plastically deformed ophiolitic peridotites (Kraka massifs, Southern Urals, Russia). *Mineralogy and Petrology* 115, 411–430. <https://doi.org/10.1007/s00710-021-00748-w>.
33. Chayka, I.F., Sobolev, A.V., Izokh, A.E., Batanova, V.G., Krashennnikov, S.P., Chervyakovskaya, M.V., Kontonikas-charos, A., **Kutyrev, A.**, Lobastov, B., Chervyakovskiy, V., 2020. Fingerprints of Kamafugite-Like Magmas in Mesozoic Lamproites of the Aldan Shield: Evidence from Olivine and Olivine-Hosted Inclusions. *Minerals* 10, 1–30. <https://doi.org/10.3390/min10040337>.
34. Stepanov, S.Y., Shagalov, E.S., Palamarchuk, R.S., **Kutyrev, A.V.**, Sharpenok, L.N., Nabiullin, F.M., Troshkina, A.N., 2020. Mineralogical and geochemical evidence for granite metasomatism in dikes in the north of the Berezovskoe Ore field (Middle Urals). *Russian Geology and Geophysics* 61, 809–827. <https://doi.org/10.15372/RGG2019176>.
35. Zhegunov, P.S., **Kutyrev, A.V.**, Zhitova, E.S., Moskaleva, S.V., Schweigert, P.E., 2024. First data on the mineralogy of the Evevpena epithermal silver–gold ore occurrence, Kamchatka, Russia. *Journal of Volcanology and Seismology* 18, 149-164. <https://doi.org/10.1134/s0742046324700489>

Under review:

36. **Kutyrev, A.**, Maier, W., Karinen, T., Barnes, S.-J., Hayes, B., Savard, D., Virtanen, V., Booth, C., Muir, D. Petrogenesis of the Akanvaara layered intrusion, Finnish Lapland. *Journal of Petrology*, under review.
37. Chayka, I., Hu, W.-J., **Kutyrev, A.**, Petrenko, D., Zhitova, L., Shevko, A., Gora, M., Kalugin, V., Shvedov, G., Izokh, A., Yudovskaya, M. Fluid-mediated PGE enrichment of magmatic low-sulfide ores as modelled for the Norilsk deposits. *Geology*, under review.

Published code and datasets:

38. Filimonova, O., **Kutyrev, A.**, & Wermeille, D., 2025. Iridium  $L_3$ -edge XAS dataset of natural Ir-O phase,

comprising  $\text{IrO}_2$  [Dataset]. Zenodo. <https://doi.org/10.5281/zenodo.15650871>

39. **Kutyrev, A.V.**, Maier, W.D., Karinen, T., Barnes, S.J., Savard, D., 2025. Whole-rock geochemical dataset of the Akanvaara layered intrusion, Lapland, Finland. Interdisciplinary Earth Data Alliance (IEDA). <https://doi.org/10.60520/IEDA/113569>. (Under embargo as per November 27<sup>th</sup>, 2025)
40. Chayka, I. F., Kamenetsky, V. S., **Kutyrev, A. V.**, Baykov, N. I., Barnes, S., Savard, D., Alexei, I. V., 2025. Major, trace, and platinum-group elements in the Late Cretaceous arc volcano-plutonic complex of the Tumrok Range, eastern Kamchatka, Version 1.0. Interdisciplinary Earth Data Alliance (IEDA). <https://doi.org/10.60520/IEDA/113578>. Accessed 2025-07-06.
41. **Kutyrev, A.**, Bindeman, I., O'Neil, J., Rizo, H., 2024. Code for 1- and 2-stage oxygen fractionation and metamorphic dehydration modeling from "Seawater-oceanic crust interaction constrained by triple oxygen and hydrogen isotopes in rocks from the Saglek-Hebron complex, NE Canada", zenodo.org. <https://doi.org/10.5281/zenodo.12795578>.

#### Conference talks:

42. IMSG-2026, Barberton, South Africa, oral presentation "Mottled anorthosites and discordant ultramafic pipes in the MG4-UG1 interval of the Upper Critical Zone, Bushveld Complex".
43. Geocongress-2025, Bloemfontein, South Africa, oral presentation "Petrogenesis of the Akanvaara layered intrusion".
44. MDSG-2024, Edinburgh, poster "Assessing mass-balance of layered intrusions: the Akanvaara intrusion case study in Lapland, Finland".
45. Goldschmidt 2023, Lyon, France, remote oral talk "Early Earth's hydrosphere: insights from the hydrothermally altered submarine lavas of the Eoarchean Saglek-Hebron Complex, Canada", 2023.
46. 14th International Platinum Symposium, Cardiff, UK, in-person oral presentation "Origin of native osmium in metamorphosed peridotites: insights from multiphase inclusions (Adamsfield placer, Tasmania)", 2023.
47. Virtual Early Career Platinum Symposium 2022, online conference, oral presentation "Noble metals in arc basaltic magmas worldwide: a case study of modern and pre- historic lavas of the Tolbachik volcano, Kamchatka".
48. EGU General Assembly 2021, online conference, oral presentation "Awaruite ( $\text{Ni}_3\text{Fe}$ ) as a platinum-group elements concentrator: Preliminary data".
49. SGA 2019 Conference, Glasgow, United Kingdom, oral in-person presentation "Platinum mineralization in Ural-Alaskan complexes of Koryak Highlands (far east Russia)".